



Network Security CS3403

Lab 6

Introduction to Uplinks between Switches

Lab 6: Focus

- Introduction to Uplinks
- Lab Configuration
 - Ping between Default VLANs
 - Ping between VLAN 10
 - Ping between VLAN 10 and VLAN 20

Course page where the course materials will be posted as the course progresses:



Introduction to Uplinks

Uplink Ports on the Switch



2 Uplink ports for interconnecting other network devices

Note: Uplink port speed is 10 Gbps whereas the Data ports are at 1 Gbps



**Uplink ports:
Ethernet 1/2/1
and Ethernet 1/2/2**

12 Downlink ports to connect to local hosts

- **Uplink ports:** when two switches are connected via their uplink ports, they are typically on the same Local Area Network (LAN).
- A connection between the uplink ports creates a single broadcast domain, enabling devices connected to either switch to communicate with each other.

Uplink Ports: Explained

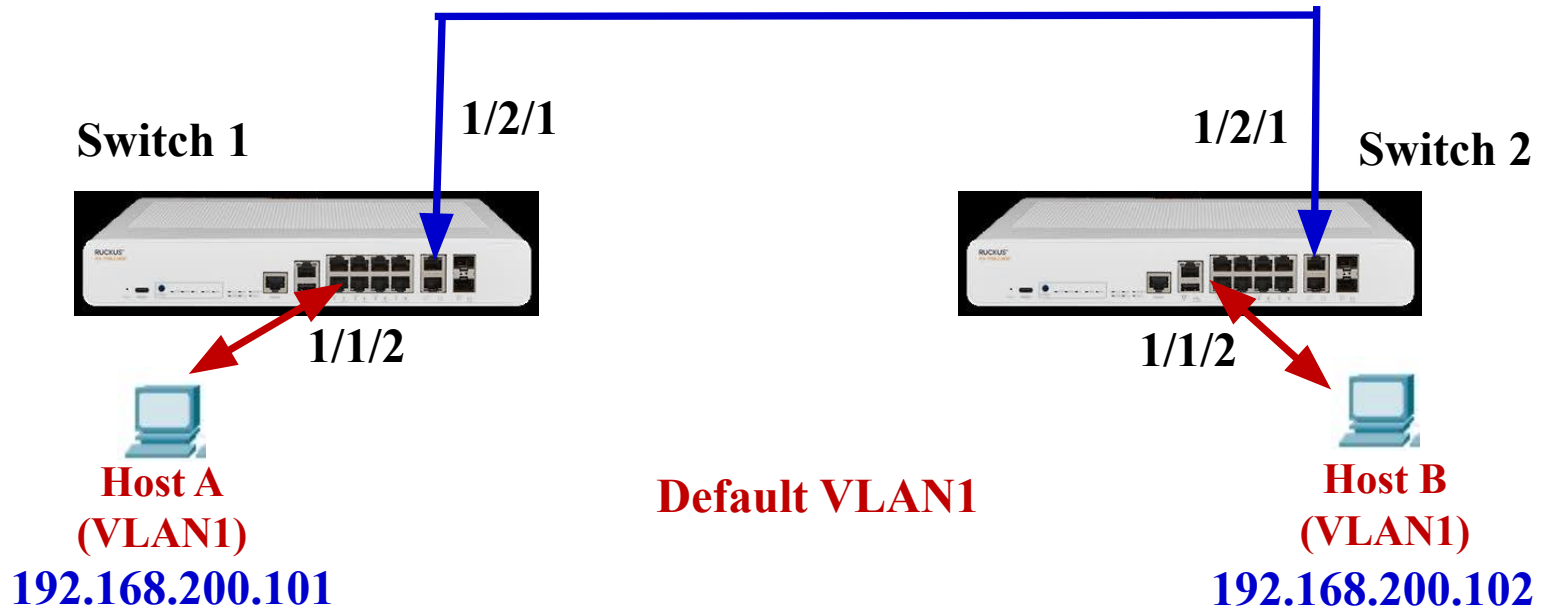
- **Broadcast Domain:** A LAN is essentially a broadcast domain. This means that when a device sends a broadcast message, all other devices on that network (within the same broadcast domain) will receive it.
- **Uplink Ports:** When two switches are connected via their uplink ports, they effectively extend the broadcast domain of the first switch to include the second switch and all devices connected to it.
- **Functionality:** This setup allows devices on either switch to communicate directly with devices on the other switch without needing to go through a router or other device.
- **Spanning Tree Protocol (STP):** In a network with multiple switches, STP is often used to prevent network loops. STP will typically disable redundant links, ensuring that there's only one path for data to flow between two switches.



Uplink Demo: Configuration

Connecting Two Switches through Uplink ports

Uplink Connection



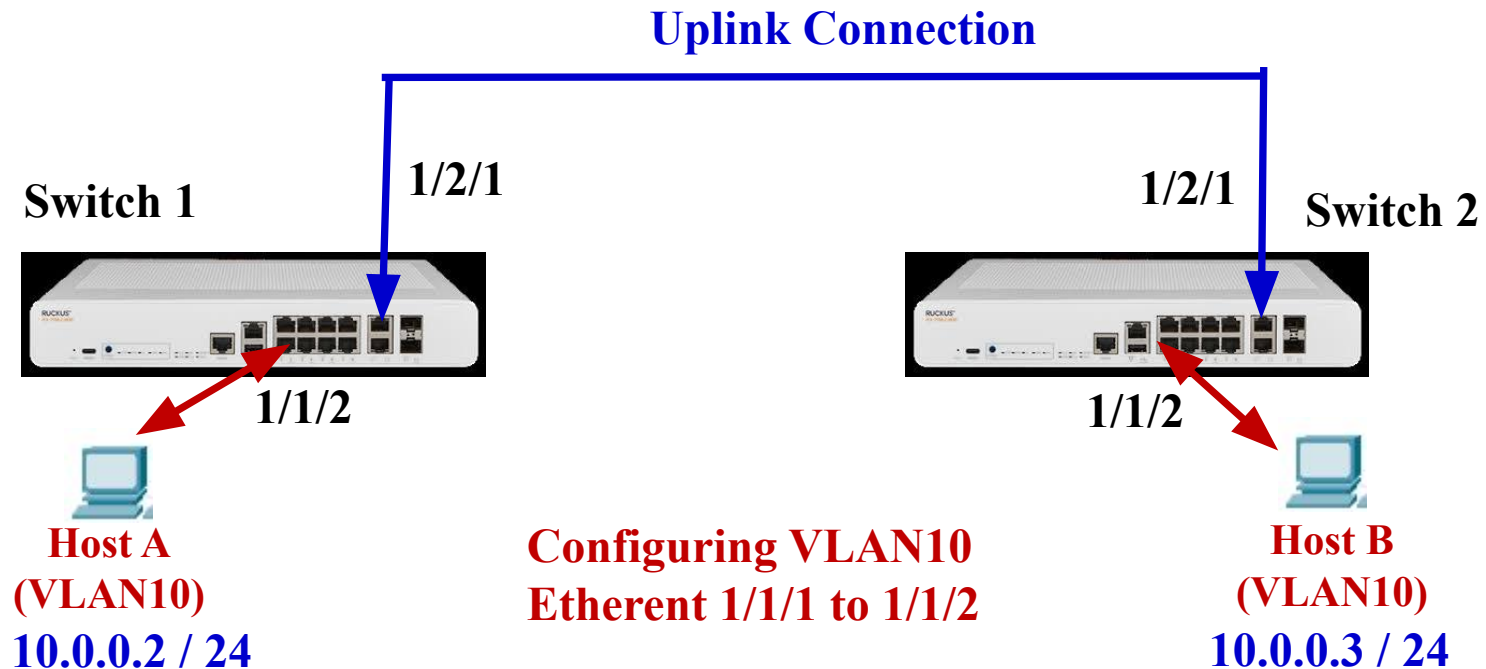
These two desktops can ping each other because Default VLAN1 of both the switches are on the same broadcast domain

```
C:\Users\Mouli>ping 192.168.200.101

Pinging 192.168.200.101 with 32 bytes of data:
Reply from 192.168.200.101: bytes=32 time<1ms TTL=128
Reply from 192.168.200.101: bytes=32 time<1ms TTL=128
Reply from 192.168.200.101: bytes=32 time<1ms TTL=128
Reply from 192.168.200.101: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.200.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Configuring VLAN 10 on both Switches



- Configure on both the switches VLAN 10 and configure the IP addresses of the desktops as shown above.
- To make these two desktops to ping each other, the **uplink port Ethernet 1/2/1 on both the switches** should be **configured** to **allow tagged VLAN 10** traffic to **pass through it**.

Configure VLAN 10

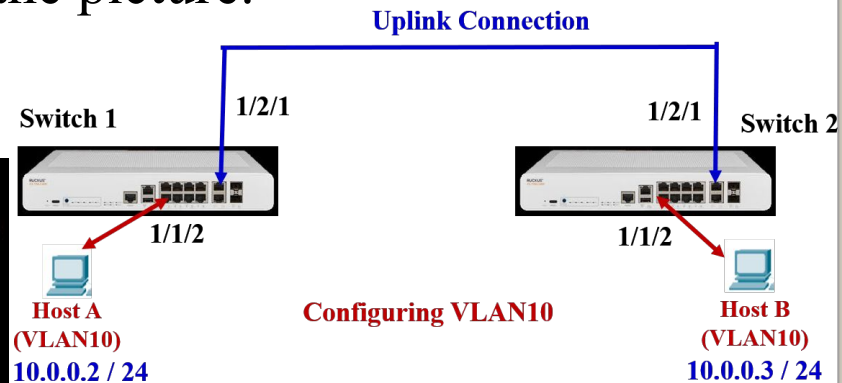
Note: Give the below commands on both the switches.

- On both the switches give the below commands
- You should be able to ping the desktops from each other once the desktops IP addresses are configured as shown in the picture.

VLAN10 (untagged): Ethernet 1/1/1 to 1/1/2

VLAN10 (tagged): Ethernet 1/2/1

```
ICX7150-C10ZP Switch(config)#
ICX7150-C10ZP Switch(config)#vlan 10 name VLAN10
ICX7150-C10ZP Switch(config-vlan-10)#untagged ethernet 1/1/1 to 1/1/2
Added untagged port(s) ethe 1/1/1 to 1/1/2 to port-vlan 10.
ICX7150-C10ZP Switch(config-vlan-10)#tagged ethernet 1/2/1
Added tagged port(s) ethe 1/2/1 to port-vlan 10.
ICX7150-C10ZP Switch(config-vlan-10)#exit
ICX7150-C10ZP Switch(config)#
```



C:\Users\Mouli>ping 10.0.0.3

From: Host A

Pinging 10.0.0.3 with 32 bytes of data:

```
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
```

Ping statistics for 10.0.0.3:

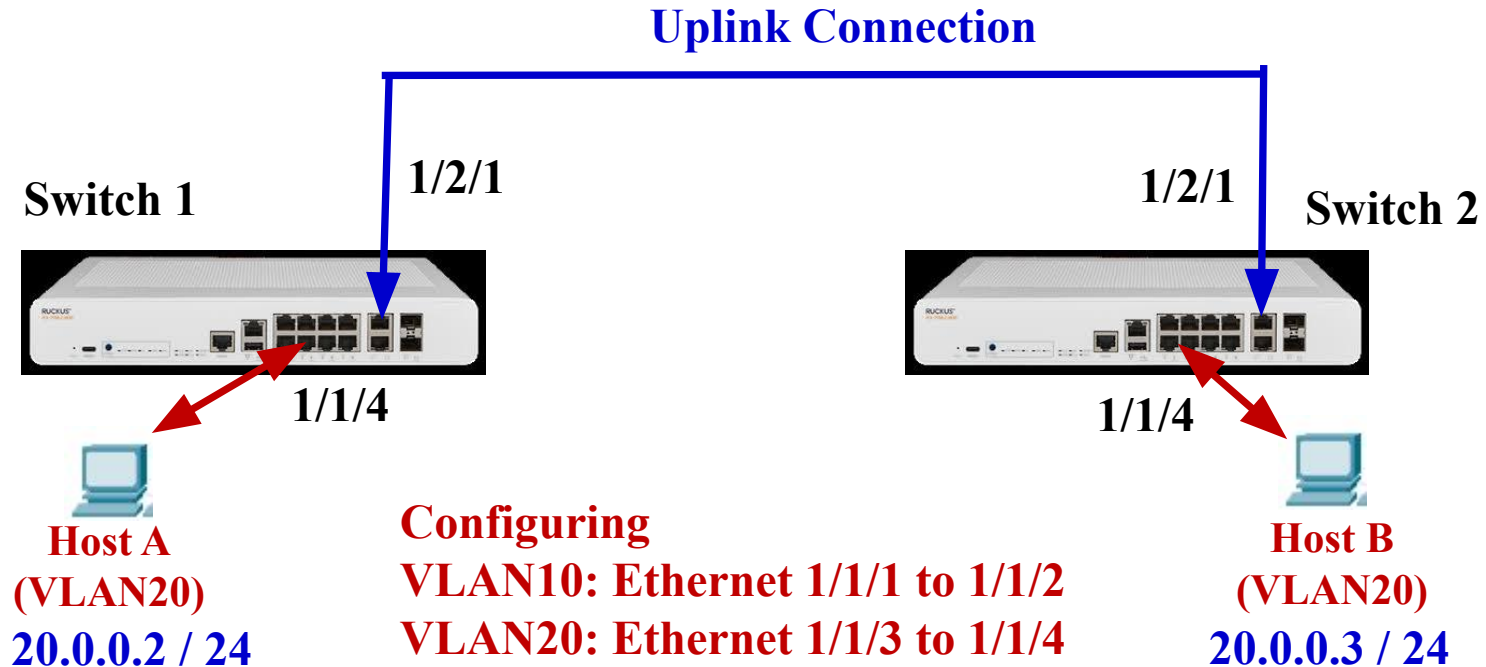
```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
ICX7150-C10ZP Switch(config)#
ICX7150-C10ZP Switch(config)#show vlan
Total PORT-VLAN entries: 2
Maximum PORT-VLAN entries: 1024

Legend: [Stk=Stack-Id, S=Slot]

PORT-VLAN 1, Name DEFAULT-VLAN, Priority level0, On
  Untagged Ports: (U1/M1) 3 4 5 6 7 8
  Untagged Ports: (U1/M2) 1 2
  Untagged Ports: (U1/M3) 1 2
  Tagged Ports: None
  Mac-Vlan Ports: None
  Monitoring: Disabled
PORT-VLAN 10, Name VLAN10, Priority level0, On
  Untagged Ports: (U1/M1) 1 2
  Tagged Ports: (U1/M2) 1
  Mac-Vlan Ports: None
  Monitoring: Disabled
ICX7150-C10ZP Switch(config)#
```

Configuring VLAN 10 and VLAN 20 on both Switches



- Apart from VLAN 10, add another VLAN 20 to both the switches.
- To make these two desktops to ping each other, the **uplink port Ethernet 1/2/1** on **both the switches** should be **configured** now to **allow tagged VLAN 20** traffic also to **pass through it**.

Configure another VLAN 20

Note: Give the below commands on both the switches.

- On both the switches give the below commands
- You should be able to ping the desktops from each other once the desktops IP addresses are configured as shown in the picture.

VLAN20 (untagged): Ethernet 1/1/3 to 1/1/4

VLAN20 (tagged): Ethernet 1/2/1

```
ICX7150-C10ZP Switch(config)#
ICX7150-C10ZP Switch(config)#vlan 20 name VLAN20
ICX7150-C10ZP Switch(config-vlan-20)#untagged ethernet 1/1/3 to 1/1/4
Added untagged port(s) ethe 1/1/3 to 1/1/4 to port-vlan 20.
ICX7150-C10ZP Switch(config-vlan-20)#tagged ethernet 1/2/1
Added tagged port(s) ethe 1/2/1 to port-vlan 20.
ICX7150-C10ZP Switch(config-vlan-20)#
ICX7150-C10ZP Switch(config-vlan-20)#exit
```

C:\Users\Mouli>ping 20.0.0.3

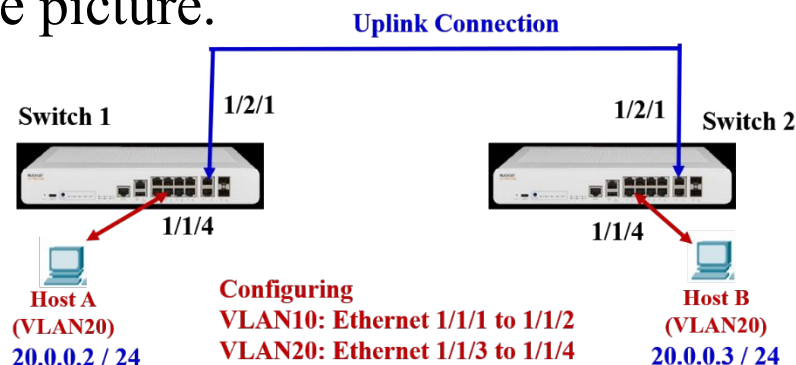
From: Host A

Pinging 20.0.0.3 with 32 bytes of data:

```
Reply from 20.0.0.3: bytes=32 time<1ms TTL=128
Reply from 20.0.0.3: bytes=32 time<1ms TTL=128
Reply from 20.0.0.3: bytes=32 time<1ms TTL=128
Reply from 20.0.0.3: bytes=32 time<1ms TTL=128
```

Ping statistics for 20.0.0.3:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
```



Configuring
VLAN10: Ethernet 1/1/1 to 1/1/2
VLAN20: Ethernet 1/1/3 to 1/1/4

```
ICX7150-C10ZP Switch(config)#
ICX7150-C10ZP Switch(config)#show vlan
Total PORT-VLAN entries: 3
Maximum PORT-VLAN entries: 1024

Legend: [Stk=Stack-Id, S=Slot]

PORT-VLAN 1, Name DEFAULT-VLAN, Priority level0, On
Untagged Ports: (U1/M1) 5 6 7 8
Untagged Ports: (U1/M2) 1 2
Untagged Ports: (U1/M3) 1 2
Tagged Ports: None
Mac-Vlan Ports: None
Monitoring: Disabled
PORT-VLAN 10, Name VLAN10, Priority level0, On
Untagged Ports: (U1/M1) 1 2
Tagged Ports: (U1/M2) 1
Mac-Vlan Ports: None
Monitoring: Disabled
PORT-VLAN 20, Name VLAN20, Priority level0, On
Untagged Ports: (U1/M1) 3 4
Tagged Ports: (U1/M2) 1
Mac-Vlan Ports: None
Monitoring: Disabled
ICX7150-C10ZP Switch(config)#
```

Lab 6: Summary

- Introduction to Uplinks
- Lab Configuration
 - Ping between Default VLANs
 - Ping between VLAN 10
 - Ping between VLAN 10 and VLAN 20